

Does Adding Heart Signals to Brain Signals Help Predict Mental Effort?

NeuroPredict: A Complete, Honest, Externally Tested Study of EEG + ECG

Final Research Report

Written by a second-year undergraduate student

Abstract

I wanted to answer one simple question: if I use heart signals (ECG) together with brain signals (EEG), can I tell better whether a person is resting or doing a mental task, compared to using EEG alone? Many papers say yes, but most of them test on the same dataset they trained on. So I built a full pipeline, trained it on one public dataset (EEGMAT, 36 people, mental arithmetic), and then tested it on a completely different one (ds003838, 65 people, a memory task, different lab, different EEG reference) that the model never saw.

On the new dataset, the EEG + ECG ensemble was better than EEG-only by **7.7 percentage points** (64.6% vs 56.9%, $p = 0.0016$, 95% CI +3.8 to +12.3). Inside the training dataset it won in **5 out of 5** random seeds (76.1% vs 70.6%). But the story is not one-sided. EEG-only had a higher AUC (0.806 vs 0.763), although that difference is not confirmed by the confidence interval. Also, the ECG-only model (60.0%) was already about as good as EEG-only, and the ensemble was not clearly better than ECG alone. In plain words: the heart signal did a lot of the work, and the EEG model did not transfer well to new people.

Along the way I found and fixed several mistakes in my own pipeline (no fixed seed, mismatched windows, mismatched reference, a channel-count bug, and some wrong or weak statements in my first draft). I list all of them, because I think that is part of the research and not something to hide.

1. Introduction

1.1 What problem am I looking at?

When a person thinks hard, their body reacts. The brain's electrical activity changes (this is measured with EEG), and the heart also changes: heart rate goes up, and the small beat-to-beat timing changes (this is called heart rate variability, or HRV, measured with ECG). If a computer can read these signals and tell whether someone is relaxed or mentally busy, it can be useful in areas like driver alertness, wearable stress trackers, learning apps, and safety-critical jobs.

1.2 Why EEG + ECG together?

Most systems use only EEG. But the brain and heart are connected through the body's automatic nervous system, so the heart carries extra information about mental effort. Adding ECG to EEG is not a new idea. Other papers have tried it. My contribution is not the idea itself but how carefully I tested it.

1.3 What bothered me about existing work

Many papers train and test on the same dataset: same lab, same machine, same task, maybe only different people. That is an easy test. A real product meets new people, new machines and new situations. So I decided to test on a completely different dataset, with a different task, to see if any improvement really holds up.

1.4 My research question

Does an EEG + ECG ensemble generalize better than EEG-only when tested on data the model has never seen, meaning different people, a different lab and a different task?

1.5 Words used in this report

Word	What it means (simple)
EEG	Electrical signals from the brain, measured with sensors on the scalp.
ECG	Electrical signals from the heart.
Heart rate (HR)	How many times the heart beats per minute.
HRV, SDNN, RMSSD	How much the time between heartbeats changes. SDNN and RMSSD are two ways to measure that.
Rest / Task	The two states I try to predict: sitting quietly vs doing a mental task.
Balanced accuracy	Average of 'how often rest is predicted correctly' and 'how often task is predicted correctly'. 50% is like guessing.
AUC	How well a model can rank task above rest, without choosing a fixed cut-off. 0.5 is guessing, 1.0 is perfect.
Ensemble	Combining two models by averaging their answers.
Leave-one-subject-out (LOSO)	Train on 35 people, test on the 1 left out, and repeat for everyone. The tested person is never seen in training.
Data leakage	When information from the test data secretly reaches training, making results look better than they are.
Seed	A number that fixes the randomness in training so the run can be repeated.
Confidence interval (CI)	A range in which the true value probably lies. If a CI for a difference does not include 0, the difference is probably real.
p-value	How surprising the result would be if there were really no effect. Below 0.05 is usually called significant.
Cohen's d	How big a difference is. About 0.2 is small, 0.5 medium, 0.8 large.
Bootstrap	Re-sampling the same data many times to see how much a number wobbles.
Calibration	Whether the model's probabilities are 'centred' properly, so a cut-off of 0.5 makes sense.

2. Aim and Objectives

Main aim: to test honestly whether combining EEG and ECG predicts rest vs mental task better than EEG alone, on data that was never used in training.

Objectives:

1. Study the training data first (EDA) to check that the task really changes the brain and heart signals.
2. Build an EEG model, an ECG model, and a way to combine them.
3. Build the pipeline so no test information leaks into training.
4. Repeat training with different seeds to see if the result is stable.
5. Test on a completely different dataset (different people, lab, task and EEG reference).
6. Report everything, including the parts that did not work or make the story messy.

3. What I Claim and What I Do Not Claim

I am writing this before the results because student projects often let the results section make the claim sound bigger than it is.

I claim	I do NOT claim
On the new dataset, the ensemble beats EEG-only in balanced accuracy at the normal 0.5 cut-off (+7.7 points, $p = 0.0016$).	That the ensemble ranks better in general. EEG-only has the higher AUC (0.806 vs 0.763), though that gap is not confirmed.
Inside the training dataset, the ensemble beats EEG-only in all 5 random seeds.	That the ensemble beats ECG-only. On the new dataset this was not confirmed (CI includes 0).
The improvement is seen across 65 new people, and the ensemble never did worse than EEG-only for any of them.	That a plain weighted average is the best way to combine the signals. A stacked model scored higher (70.0%).
The EEG model transferred poorly to new data, and the heart-rate-based model was about as good.	That this works everywhere. I have only one successful external dataset, and a second one was rejected because it did not fit.
The whole pipeline is repeatable: settings are checked automatically and long runs are saved in checkpoints.	That this is ready for real-world use. 64.6% is far too low for a safety product.

4. Data

4.1 Training data: EEGMAT (PhysioNet)

36 people. Each person has one recording at rest and one recording while doing mental arithmetic (subtracting numbers in the head). Each file has EEG and ECG together. Each file has 21 channels: 19 scalp EEG channels, one ear channel (A2-A1) and the ECG channel. I used the standard 19 scalp channels of the 10-20 system: Fp1, Fp2, F3, F4, F7, F8, T3, T4, C3, C4, T5, T6, P3, P4, O1, O2, Fz, Cz, Pz. Sampling rate is 500 Hz. The original reference is 'linked ears' (I checked this in the PhysioNet documentation, not by guessing).

4.2 External test data: ds003838 (OpenNeuro)

65 people (out of a bigger group) who have both usable EEG and ECG. The task is different: a digit-span working-memory task (remember a sequence of numbers). The EEG has 63 channels and the reference is FCz. I picked this on purpose. Testing on a different task is a harder and more honest test than testing on the same task with new people.

4.3 The two datasets side by side

	EEGMAT (training)	ds003838 (external test)
Source	PhysioNet	OpenNeuro
People	36	65 (all with usable EEG and ECG)
Rest condition	Resting recording	Resting recording (I use the part before the 'eyes opened' marker)
Task condition	Mental arithmetic	Digit-span memory task
EEG channels in file	21 (19 scalp + 1 ear channel + ECG)	63
Channels I used	19 standard channels	The same 19. T7, T8, P7, P8 are the modern names for T3, T4, T5, T6, so I treated them as the same positions.
Original reference	Linked ears	FCz
Reference I used for both	Average reference	Average reference
Sampling rate	500 Hz	Resampled to 500 Hz
Role in this project	Train and internal check	Final, never-seen test

4.4 A dataset I tried and rejected

I also looked at MPD-DF, a driving-fatigue dataset with EEG and ECG (50 people), hoping for a second external test. But there, 'Wakefulness' means the driver is actively driving (closer to my 'task'), while 'Fatigue' means the driver is drifting towards sleep, which is less engagement, not more. Forcing this into my rest-vs-task labels would have given labels a meaning they do not have. So I did not use it. Knowing when a dataset does not fit is part of doing research properly.

5. First Look at the Data (EDA)

Before building any model, I checked if the mental task really changes the body. I extracted EEG band power and ECG heart features for all 36 people in both conditions. Everything worked: **72 out of 72** recordings (36 people x 2 conditions) gave results, with no failures.

5.1 EEG band power: rest vs task

Band power means how strong the brain signal is in a certain speed range (delta is slowest, gamma is fastest). I averaged over all channels for each person, then ran a paired t-test for each band.

Band	Change in task vs rest	t value	p value	Significant?
Delta	+11.88%	-1.100	0.27864	No
Theta	-14.24%	2.759	0.00916	Yes
Alpha	-33.85%	4.665	0.00004	Yes
Beta	-12.80%	4.392	0.00010	Yes
Gamma	+19.40%	-3.232	0.00268	Yes

Table 1. t is calculated as rest minus task, so a positive t means the band dropped during the task. Four bands stay significant even with a simple Bonferroni correction for 5 tests (cut-off 0.01).

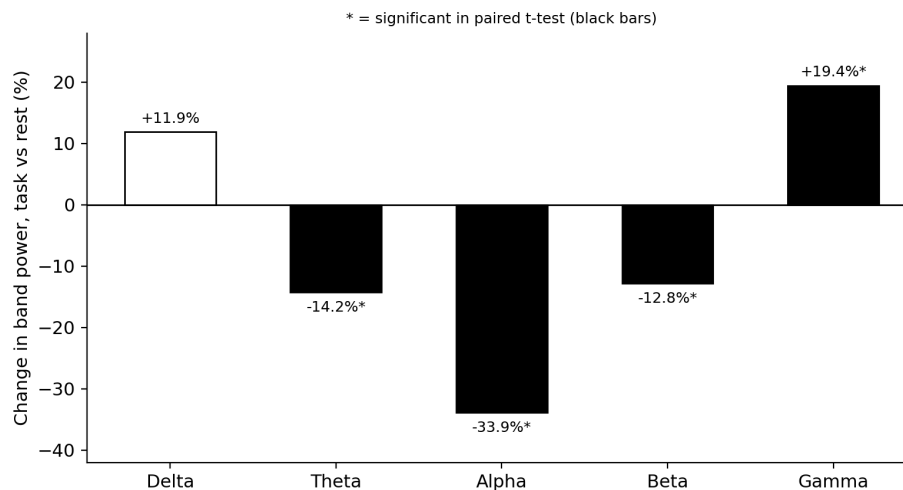


Figure 1. Change in EEG band power during the task. The biggest change is the alpha drop.

The famous expected pattern is that alpha falls over the back of the head (occipital and parietal) during a mental task. I saw **-37.41%** there, and **33 out of 36 people** showed this drop. I also expected frontal theta to rise; it rose by only **+7.35%**. Theta over all channels actually went down, so the theta story is mixed and I do not push it. The gamma increase could partly be muscle noise, but I did not test that.

5.2 ECG: heart rate and HRV

Feature	Rest (mean, SD)	Task (mean, SD)	Paired t-test p
Heart rate (bpm)	73.85 (13.73)	86.57 (18.89)	0.00022 (significant)
SDNN (ms)	141.99 (303.38)	125.61 (232.57)	0.80481 (not significant)
RMSSD (ms)	173.07 (443.45)	135.60 (273.50)	0.67703 (not significant)

Table 2. ECG features for all 36 people.

Heart rate rose by about 13 bpm during the task, which is what I would expect if the task really worked on the body. But SDNN and RMSSD did not change significantly, and their spread is huge (SD of 300 to 440 ms). Normal values are far smaller, so some recordings probably have wrong beat detection. This matters later: **my ECG model is in practice mostly a heart-rate model.**

5.3 Do brain change and heart change go together?

I checked if people whose alpha dropped more also had a bigger heart-rate rise. The correlation was $r = 0.071$, $p = 0.679$, so no link. This does not prove the two signals carry different information, but it fits the idea that they may add something different to each other.

5.4 What the EDA taught me

- The task really changes both brain and heart signals, so the labels mean something real.
- Alpha suppression at the back of the head is the clearest brain signal (33 of 36 people).
- Heart rate is the clean ECG signal; SDNN and RMSSD are noisy.

6. Methods

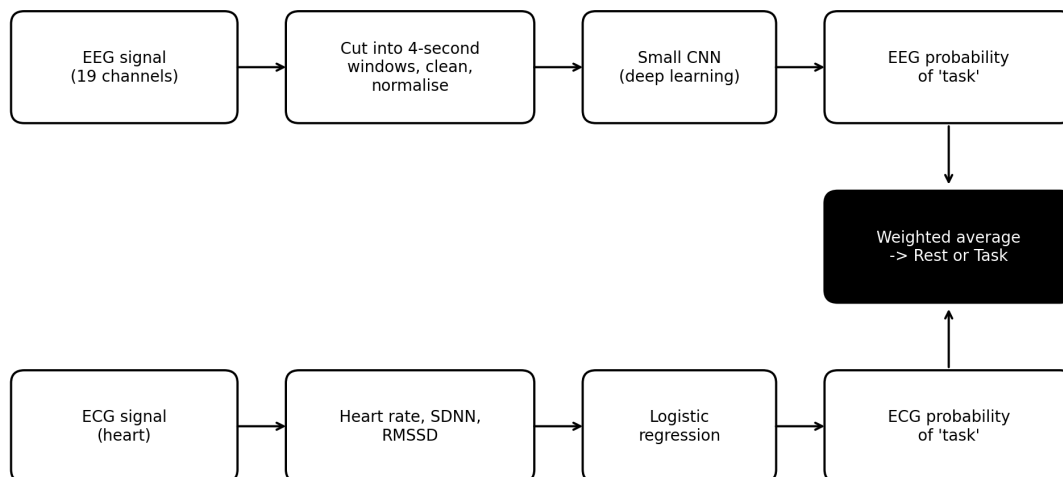


Figure 2. Overall pipeline. EEG and ECG each make their own probability of 'task', then a weighted average decides.

6.1 EEG preparation

- Band-pass filter 1 to 40 Hz to remove slow drifts and high-frequency noise.
- Average reference (needed because the two datasets use different references).
- Cut into 4-second windows with a 2-second step. Each window is 19 channels x 2000 samples. EEGMAT gave 4,266 windows in total.
- Each window is normalised (z-score per channel).

6.2 EEG model

A small convolutional neural network (CNN) with three blocks. Each block has a 1D convolution (32, 64, 128 filters), batch normalisation, ReLU and max pooling, followed by dropout (0.5), global average pooling and a final layer with 2 outputs. In my code the class is called EEGNet, but it is my own small design, not the published EEGNet model. Training: 15 epochs, Adam optimiser (learning rate 0.0001, weight decay 0.0001), batch size 32, class-weighted loss and gradient clipping.

6.3 ECG model

Three hand-made features per recording: mean heart rate, SDNN and RMSSD. They are scaled and fed to a logistic regression (a very simple model) with balanced class weights.

6.4 Ensemble

Final probability = $W_{\text{EEG}} \times (\text{EEG probability}) + W_{\text{ECG}} \times (\text{ECG probability})$. Each weight is that model's accuracy divided by the sum of both accuracies, so the better model gets a bit more say. All weights come from training-side accuracy only. In the internal test, weights were worked out separately in every fold from a small inner validation group of training people. For the external test I used fixed weights $W_{\text{EEG}} = 0.495$ and $W_{\text{ECG}} = 0.505$.

6.5 Internal test: leave-one-subject-out, 5 seeds

For each of the 36 people: train on the other 35, test on the one left out. I repeated this whole process with 5 different seeds (36 folds x 5 seeds = 180 training runs, saved in a checkpoint file so a crash would not lose progress). For each person and condition I averaged the window probabilities and then used the 0.5 cut-off. That gives 72 test points per seed.

6.6 External test on ds003838

I trained one final EEG model and one final ECG model on all 36 EEGMAT people and saved them with a settings file. On the new dataset the notebook automatically checks that window length, step, channel count, sampling rate and reference match the training settings, and stops with an error if anything differs. Then, for each of the 65 people:

- **EEG:** rest windows slide over the rest recording (4 s, step 2 s). Task windows start at each memory-trial onset (4 s each). The model gives a probability per window, then I average them.
- **ECG:** heart beats are found with NeuroKit2. Beat gaps outside 300 to 2000 ms are dropped. Rest uses the whole rest segment. Task uses memory runs pooled up to 200 seconds, so both conditions have similar lengths. At least 60 seconds is needed.
- Each person gets one rest point and one task point, so every score can only be 0, 0.5 or 1.

6.7 Statistics I used

- Bootstrap 95% confidence intervals (2,000 re-samples over people).
- Wilcoxon signed-rank test for paired differences per person.
- Cohen's d for effect size.
- ROC-AUC, confusion matrices, and an ablation of three ways to combine the signals.

7. Making Sure I Was Not Fooling Myself

This is the part I spent most time on. Some mistakes I made during the work, and some I found later while re-checking my report line by line against my notebooks. Both kinds are listed.

What went wrong	How I noticed	How it is fixed (or its status)
No fixed random seed	Same code gave 75.0% in one run and 69.4% in another	Fixed 5 seeds. I report mean and spread instead of one lucky number.
Window length mismatch	External test used 2 s windows but the model was trained on 4 s	Both use 4 s window and 2 s step. The notebook now checks this automatically.
EEG reference mismatch	Read both datasets' documents: linked ears vs FCz	Both re-referenced to average reference; the notebook checks the reference too.
Channel-count bug	A first clean run crashed: class default was 21 channels, data had 19	The real channel count is passed everywhere.
Ensemble weights from an older training accuracy	Found while writing the first draft: 0.6667 used, corrected value is 0.7056	Still open. I could not re-run the external test here. The corrected weights would be about 0.509 / 0.491 instead of 0.495 / 0.505. The ablation gives a hint that this is minor: a plain 50/50 average scored 63.8% vs 64.6% for my weights. Re-run is listed in Future Work.
Report said 74 of 74 recordings worked	Checked against the EDA notebook: 36 x 2 = 72 files	Corrected to 72 of 72.

What went wrong	How I noticed	How it is fixed (or its status)
A test that proves nothing: '65/65 people scored at least 50%, $p < 0.0001$ '	EEG always gets 'task' right, so its score can never fall below 0.5. The test cannot fail.	Dropped from the report as evidence.
Effect size called 'medium'	$d = 0.423$ is between small (0.2) and medium (0.5)	Now called small-to-medium.
AUC gap described too strongly	The CI for the AUC difference is -0.095 to +0.007, which includes 0	Now described as: EEG-only has a higher AUC number, but the gap is not confirmed.
Ensemble was never compared with ECG-only	Bootstrap output existed but was not discussed	Added in Section 8.4. This changes the story (see there).
Only 19 of 63 channels used on the new dataset, and T7/T8/P7/P8 treated as T3/T4/T5/T6	Read the external notebook	Stated openly in Section 4.3. It is needed because the model expects the same 19 inputs.
Model called EEGNet	The class name is EEGNet but the design is my own	Now called a small custom CNN.

Table 3. Problems found and what happened to each.

8. Results

8.1 Internal test (EEGMAT, 5 seeds)

Seed	EEG-only	ECG-only	Ensemble	Ensemble minus EEG
1	69.4%	68.1%	75.0%	+5.6
2	73.6%	68.1%	77.8%	+4.2
3	68.1%	68.1%	73.6%	+5.6
4	69.4%	68.1%	76.4%	+6.9
5	72.2%	68.1%	77.8%	+5.6
Mean	70.6%	68.1%	76.1%	+5.6 (SD 1.0)

Table 4. Balanced accuracy per seed. ECG-only is identical in every seed because logistic regression has no randomness in this setup.

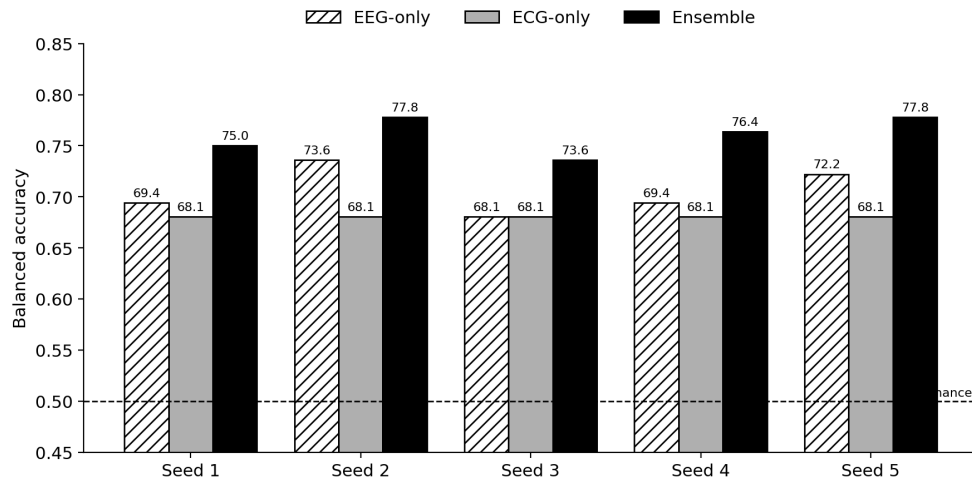


Figure 3. The ensemble beats EEG-only in every one of the 5 seeds.

The ensemble wins **5 out of 5** times, and the average gain is about 5.6 points. One honest note: the 5 seeds all use the same 36 people, so they show the result is not a fluke of training randomness, but they are not 5 independent tests on new people. The external test covers that.

8.2 External test (ds003838, 65 new people)

Model	Balanced accuracy	95% CI	AUC
EEG-only	56.9%	53.1% to 61.5%	0.806
ECG-only	60.0%	55.4% to 65.4%	0.675
Ensemble	64.6%	59.2% to 70.0%	0.763

Table 5. External results. Chance level is 50%.

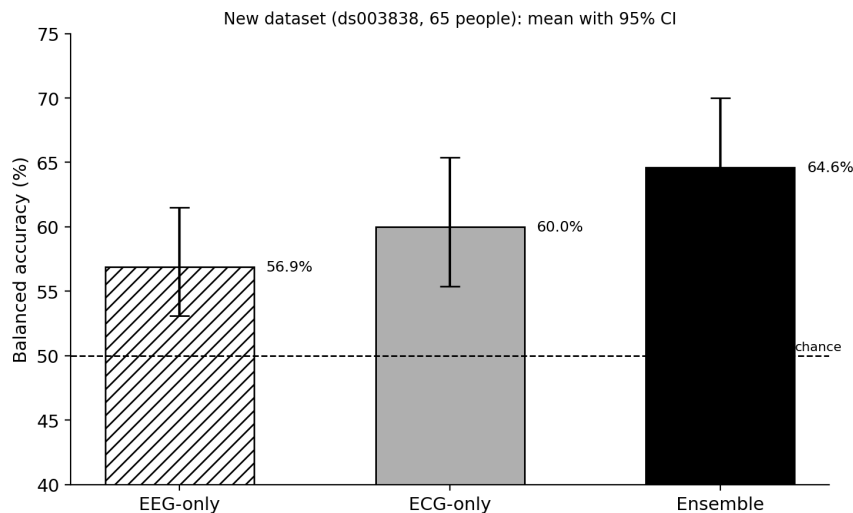


Figure 4. Accuracy on the new dataset with 95% confidence intervals.

The ensemble beat EEG-only in **10 of 65** people, tied in 55, and lost in **0**. The mean improvement was **+7.7 points**, Wilcoxon $p = 0.0016$, and the 95% CI of the improvement was **+3.8 to +12.3**, which does not include zero. Cohen's d was 0.423 (small-to-medium).

A tie does not mean nothing happened. It means the person's score stayed the same (for example EEG and ensemble both got rest wrong and task right).

8.3 Where the improvement comes from

Model	Rest correct	Task correct	Balanced accuracy
EEG-only	9 of 65 (14%)	65 of 65 (100%)	56.9%
ECG-only	43 of 65 (66%)	35 of 65 (54%)	60.0%
Ensemble	19 of 65 (29%)	65 of 65 (100%)	64.6%

Table 6. Confusion matrix summary at the 0.5 cut-off (65 people x 2 conditions).

This table explains a lot. The EEG model says 'task' for almost everyone, so it gets task right every time but rest right only 9 times. Its 56.9% is close to what you get by always answering 'task'. The ECG model makes more balanced mistakes. The ensemble pulls the EEG model's answers down towards 'rest' and fixes 10 more rest cases (9 to 19), without losing any task case. That is exactly the 10 wins.

8.4 Ensemble vs ECG-only (a comparison I had missed)

Comparison	95% CI of the difference
Ensemble minus EEG-only	+3.8 to +12.3 points (confirmed)
ECG-only minus EEG-only	-3.1 to +10.0 points (not confirmed)
ECG-only minus Ensemble	-11.5 to +2.3 points (not confirmed)

Table 7. Bootstrap comparisons on the new dataset.

So the honest reading is: the ensemble is clearly better than EEG-only, but I cannot say it is better than the heart model alone. The ECG model, which is simple and mostly uses heart rate, was already about as good. Adding ECG mainly rescued a weak EEG model.

8.5 The AUC complication

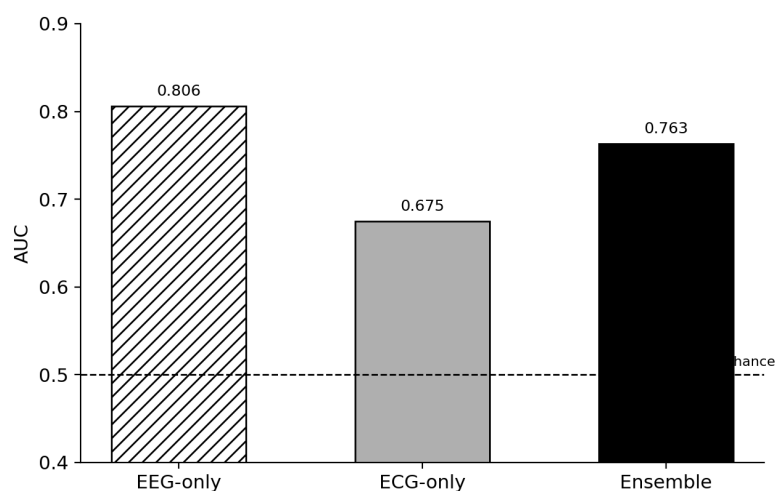


Figure 5. AUC on the new dataset.

EEG-only has the highest AUC (0.806), ensemble 0.763 and ECG-only 0.675. The bootstrap CI for ensemble minus EEG-only AUC was -0.095 to +0.007, so it includes zero: the AUC difference is not confirmed either way. My reading is that the EEG model can rank rest below task quite well, but its default cut-off of 0.5 is in the wrong place on the new dataset (it calls almost everything 'task'). The ensemble fixes part of that cut-off problem, which raises accuracy, without adding much real

new information, which is why AUC does not go up. I tried Platt scaling (re-calibrating the ECG probabilities): ECG AUC stayed at 0.675. Two further checks from my analysis notes† (re-calibration of the ensemble and weighting by AUC) also stayed around 0.76 to 0.77, still below EEG-only. So this is not a simple fix and looks like a real property of the data.

Both facts are true together: **better accuracy at a fixed cut-off, but not clearly better ranking.**

8.6 Is my way of combining the signals doing anything?

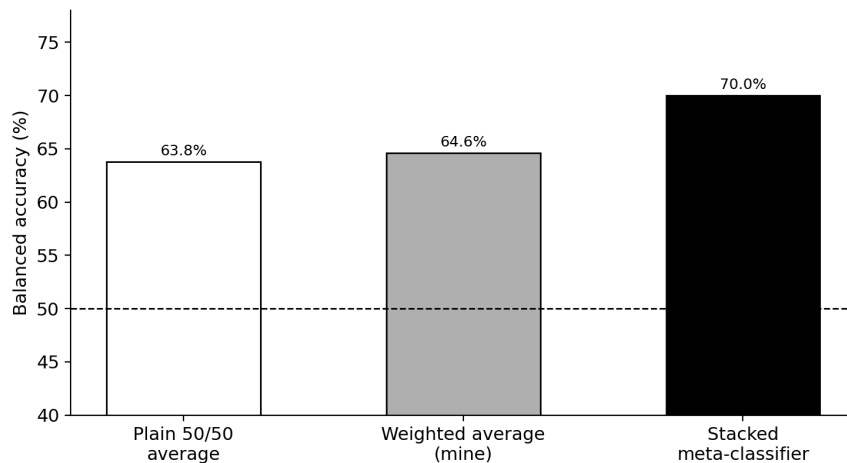


Figure 6. Three ways of combining EEG and ECG on the new dataset.

My weights are almost 50/50, so my 'confidence-weighted' average is basically a plain average (64.6% vs 63.8%). A small stacked meta-classifier (logistic regression, tested leave-one-subject-out) reached **70.0%**. I worried this was unfair, because I had tried 3 schemes on the same 65 people and reported the best, which is a known trap. So I built a stricter test†: for every person, pick the method using only the other 64 people, then apply it. The stacked model was chosen 65 out of 65 times and the honest accuracy stayed at 70.0%. So for these two schemes the advantage looks real. I only ran this stricter test for 2 of the 5 schemes I originally tried.

† Extra checks (Platt scaling on the ensemble, AUC-based weighting, nested selection) were run as additional analysis and are reported here as in my analysis notes; they are not part of the three main notebooks.

9. What I Improved Along the Way

Step	Before	After
Fixed seeds and 5 repeats	One run: 75.0% or 69.4%, depending on luck	Mean 76.1% (SD 1.8) for ensemble, 70.6% (SD 2.3) for EEG
Matching windows and reference	Silent mismatch between training and test	Same settings, checked automatically by the notebook
Adding ECG to EEG (internal)	EEG-only 70.6%	Ensemble 76.1% (+5.6)
Adding ECG to EEG (external)	EEG-only 56.9%	Ensemble 64.6% (+7.7)
Rest predicted correctly (external)	9 of 65 people	19 of 65 people

Step	Before	After
Better combination method (external)	Weighted average 64.6%	Stacked meta-classifier 70.0%
Honesty checks	Only headline numbers	CIs, effect size, AUC, confusion matrices, ablation, nested selection, bug table

Table 8. Summary of improvements.

10. What Is Different or Useful About This Project

I do not claim to have invented anything new. EEG + ECG fusion exists already. What I think is less common, especially in student work, is this combination:

- **A truly external test:** different people, a different lab, a different task and a different EEG reference, and settings that are checked automatically in code.
- **Repeating everything with 5 seeds** and reporting the spread instead of one number.
- **A written bug log** with mistakes found by me, not by a reviewer, including ones I found in my own report.
- **Reporting a result that complicates my own story** (the AUC gap and the ECG-only comparison).
- **A check against choosing the best of many methods** (nested selection).
- **Rejecting a dataset** that did not fit instead of forcing labels.
- **Resume-safe notebooks** with checkpoints for long runs, so results are reproducible.

11. Discussion

The main finding is real but modest. On truly new data, EEG + ECG beat EEG-only by 7.7 points with a solid confidence interval. But the effect is small-to-medium, and the accuracy (64.6%) is still low.

Most of the value came from the heart. The ECG model, which is basically heart rate, scored 60.0%, better than EEG-only at 56.9% (not confirmed as a real difference, but not worse either). The EEG model, which did well inside its own dataset (70.6%), lost most of that on new people. So the brain model did not carry over well.

Why did EEG transfer badly? I did not test this, so these are ideas, not findings: the task changed (arithmetic vs memory), the machines and caps are different, I used only 19 of the 63 channels, the reference is different in the original recordings, and a deep learning model trained on only 36 people can learn quirks of that one dataset.

Why does the ensemble help then? The EEG model was biased towards 'task'. The ECG model has a different kind of error, so averaging balanced things out. It helped the cut-off, not the ranking.

What the results say about method. Testing only inside one dataset would have hidden this. Inside EEGMAT everything looked good. Only the external test showed how fragile the EEG model was.

12. Real-World Meaning: How This Could Help

12.1 Where this kind of system can be used

Area	How it could help	What is still needed
Driver or pilot alertness	Warn when mental load or drowsiness gets too high.	Much higher accuracy, real driving data, very few false alarms.
Wearables and stress tracking	A watch or chest strap already has a heart sensor. Adding a brain sensor could make stress or overload detection better.	Comfortable dry EEG with few channels; testing outside the lab.
Learning apps and classrooms	Make lessons easier or harder depending on the student's mental effort.	Privacy rules, consent, and proof that it helps learning.
High-workload jobs (air traffic control, surgery)	Spot overload early and share the work.	Very reliable models, tested on real staff.
Adaptive computer interfaces	A system that slows down or simplifies when the user is overloaded.	Fast real-time processing (results within seconds).
Research tools	A quick check that a task really changed the body, like my EDA step.	Nothing big; this already works today.

Table 9. Possible uses. These are directions, not things this project has proven.

12.2 What my improvements mean in practice

What I improved	Why it matters in real life
Multi-seed repeats	A company would know if a product result is stable or just luck of one training run.
Testing on a different dataset	Real users are always 'new people on a new device'. This is the test that shows if a system will survive that.
Automatic setting checks	Silent mismatch between training and deployment (filters, sensors, reference) is a common reason real systems fail.
Adding a cheap heart sensor	Heart signals are easy and cheap to measure. Even a small gain from them, or a rescue of a weak brain model, is valuable.
Rest correct 9 to 19 of 65	Fewer wrong 'you are overloaded' alerts for people who are just relaxed.
Honest reporting of weak spots	In health and safety products, overpromising is dangerous. Knowing exactly where a model is weak is the first step to fixing it.

12.3 A realistic warning

64.6% balanced accuracy on a two-choice problem is not enough for driver safety or medical use. The realistic message of this project is: **combining body signals helps, testing on new data is essential, and the EEG part still needs a lot of work before real use.**

13. Limitations

- **Coarse score:** each person only gives 2 test points, so scores are 0, 50 or 100%. This is one reason confidence intervals are wide.

- **Weights are almost 50/50**, so the 'confidence-weighted' name is generous. They also came from an older training accuracy; the corrected re-run is not done yet.
- **Only one successful external dataset.** A second one was rejected for good reasons, so I have no second independent confirmation.
- **Effect size is small-to-medium** ($d = 0.423$). I do not want to oversell it.
- **The AUC vs accuracy disagreement is unresolved.** I tried two fixes and neither worked.
- **The nested fairness check covers only 2 of 5 schemes.**
- **ECG features are noisy.** SDNN and RMSSD have extreme values, so the ECG model mostly uses heart rate. I did not clean beat detection errors.
- **Different rest definitions.** In the new dataset, task windows start at trial onsets, while rest windows slide across the recording, and heart features for task come from pooled runs capped at 200 seconds. I matched lengths but the conditions are not perfectly parallel.
- **Only 19 channels** were used on the new dataset, and four channel names were treated as equal to older names.
- **Internal seeds are not independent** because they share the same 36 people.
- **Only two states** (rest vs task) were studied, not different levels of mental load.

14. Future Work

14.1 Quick fixes to finish the current study

Idea	Why	How I would check
Re-run the external test with corrected weights (about 0.509 / 0.491)	Removes the last known mismatch	Rerun the external notebook with the 0.7056 training accuracy; compare with 64.6%.
Extend nested selection to all 5 schemes	Fully rules out 'picked the best of many'	Choose the method for each person from the other 64 people, for every scheme.
Score every 4-second window, grouped by person	Fixes the coarse 0 / 50 / 100 score	Bootstrap over people while keeping each person's windows together.
Investigate AUC vs accuracy	Understand instead of just testing simple fixes	Choose the cut-off on training data; test small calibration using a person's own short rest recording.

14.2 Making the models better

Idea	Why	How I would check
Clean the ECG: remove wrong beats, use robust HRV, add features like pNN50 and LF/HF	SDNN and RMSSD are very noisy right now	Compare ECG-only accuracy before and after cleaning.
Try other EEG models (real EEGNet, simple band-power features with a classifier)	A big model on 36 people may overfit	Same external test for each, same seeds.
Train on several datasets together	More variety may reduce dataset-specific habits	Leave one whole dataset out as the test.

Idea	Why	How I would check
Normalise each person or recording separately (domain adaptation)	Reduces differences between machines and labs	Compare external accuracy with and without.
Use more of the 63 channels or a channel-selection method	Only 19 of 63 were used	Train on a common subset, then on a larger one; watch what improves.
Look inside the CNN (which channels and bands matter)	Check if it uses the alpha drop from Section 5	Saliency maps or removing one band at a time.

14.3 Towards real-world use

Idea	Why	How I would check
Find a second external dataset with a true rest-vs-task design	Needed for a second independent confirmation	Apply the same fit-or-reject rules I used for MPD-DF.
Wearable-friendly version: 4 to 8 forehead channels, and a wrist pulse sensor instead of ECG	Real devices cannot use 19 gel electrodes	See how much accuracy drops as sensors are removed.
Personal calibration with a few minutes of the user's own rest	Could fix the cut-off problem for each new person	Compare with and without calibration on the external data.
Predict levels (low, medium, high load) or a continuous score	Real use needs more than two states	Find datasets with graded difficulty.
Live (real-time) demo with a sliding window	Shows delay and stability in practice	Measure response time and false alarms on a live recording.
Add other signals (breathing, skin conductance, eye tracking)	More body signals may add new information	Ablation: add one signal at a time.

14.4 Next project

My next project uses the same habits (fixed seeds, external replication, honest bug log, reporting messy results) on a different question: whether different reinforcement-learning models of how people learn from reward can be told apart from EEG signals.

15. Questions Someone Might Ask

Q: Are your ensemble weights secretly using the test data?

A: No. They come only from training-side (EEGMAT) accuracy. The known issue is that they came from an older training number, not that they used test data.

Q: 7.7 points sounds small. Could it be noise?

A: The 95% CI is +3.8 to +12.3 and $p = 0.0016$, so it looks real but modest. I do not claim more than that.

Q: Then why not just use ECG alone?

A: That is a fair question. On the new dataset ECG-only was 60.0% and the ensemble 64.6%, but the difference was not confirmed (the CI for ECG-only minus ensemble was -11.5 to +2.3, which includes 0). Inside EEGMAT the ensemble was clearly higher (76.1% vs 68.1%). More data is needed to settle it.

Q: Your AUC result seems to contradict your main claim.

A: Both are true. Accuracy at a fixed cut-off improved, ranking did not clearly improve, and the AUC gap itself is not confirmed. I reported it because it complicates the clean story.

Q: Did you just pick the best of several ensemble methods?

A: At first, yes. So I built the nested test, which chose the stacked method 65 of 65 times with no drop in accuracy. I only ran it for 2 of 5 schemes.

Q: Why only 19 of 63 channels on the new dataset?

A: The model was trained on 19 channels, so it needs the same 19 as input.

16. Conclusion

I set out to test, not assume, whether combining EEG and ECG predicts mental state better than EEG alone, using a leakage-free pipeline and a truly external dataset. The answer is a real but modest **yes at a fixed cut-off** (+7.7 points, $p = 0.0016$, better in 5 of 5 seeds), and a careful **not clearly** for ranking ability (AUC) and for beating the heart model alone. The most useful things I learned are that the heart signal carried much of the value, that the EEG model did not transfer well to new people, and that only a real external test shows this.

I tried to write this the way I would want a paper to be written for me to learn from: showing my mistakes, stating what I am unsure of, and stopping my claims exactly where the evidence stops.

Appendix A: Key Numbers

Metric	EEG-only	ECG-only	Ensemble
Internal balanced accuracy (5-seed mean)	70.56%	68.06%	76.11%
Internal SD across seeds	2.28	0.00	1.81
External balanced accuracy (n = 65)	56.9%	60.0%	64.6%
External 95% CI	53.1 to 61.5	55.4 to 65.4	59.2 to 70.0
External AUC	0.806	0.675	0.763
Rest correct / task correct (out of 65)	9 / 65	43 / 35	19 / 65

Ensemble vs EEG-only (external): wins 10, ties 55, losses 0. Mean change +7.7 points. Wilcoxon $p = 0.0016$. 95% CI +3.8 to +12.3. Cohen's $d = 0.423$.

Ensemble minus EEG-only AUC: 95% CI -0.095 to +0.007. Ablation: 50/50 average 63.8%, my weights 64.6%, stacked meta-classifier 70.0%.

EDA: heart rate 73.85 to 86.57 bpm ($p = 0.00022$). Occipital/parietal alpha -37.41%, 33 of 36 people. Alpha change vs heart-rate change: $r = 0.071$, $p = 0.679$.

Appendix B: Settings Used

Item	Setting
EEG filter and reference	1 to 40 Hz band-pass, average reference
Windows	4 seconds, step 2 seconds, 500 Hz (2000 samples), 19 channels
EEG model	3 conv blocks (32, 64, 128 filters; kernels 7, 5, 3), batch norm, ReLU, max pool 4, dropout 0.5, global average pool, linear layer
EEG training	15 epochs, Adam, learning rate 0.0001, weight decay 0.0001, batch 32, class-weighted loss, gradient clip 1.0
Inner validation for weights	About 1/5 of training people held out, 10 epochs
ECG features and model	Mean HR, SDNN, RMSSD; standard scaling; logistic regression with balanced weights
Seeds	1, 2, 3, 4, 5 (final model: 42)
External ECG rules	R-peaks with NeuroKit2; keep beat gaps 300 to 2000 ms; at least 60 s; task pooled up to 200 s
External weights used	$W_{\text{EEG}} = 0.495$, $W_{\text{ECG}} = 0.505$ (corrected value would be about 0.509 / 0.491)
Bootstrap	2,000 re-samples, seed 42

Appendix C: Notebooks

Notebook	What it does
1. Weighted Ensemble v3 (Colab)	Loads EEGMAT, runs leave-one-subject-out with 5 seeds, computes accuracies, trains and saves the final EEG model, ECG model and settings file.
2. External Validation on ds003838 (Kaggle)	Downloads subjects, checks settings, gets EEG and ECG predictions for 65 people, then computes accuracy, CIs, AUC, effect size, confusion matrices, ablation and ECG calibration.
3. EEG + ECG EDA on EEGMAT (Colab)	Extracts band power and HRV for 36 people, runs t-tests, alpha check and EEG-ECG correlation, and saves the feature tables.